

Computational Fluid Dynamics II

Lab 2 Part 2: Flow around a circular cylinder

Due date: 14 July 2025

Deliverables: Group report + code + data

Report format: Typed

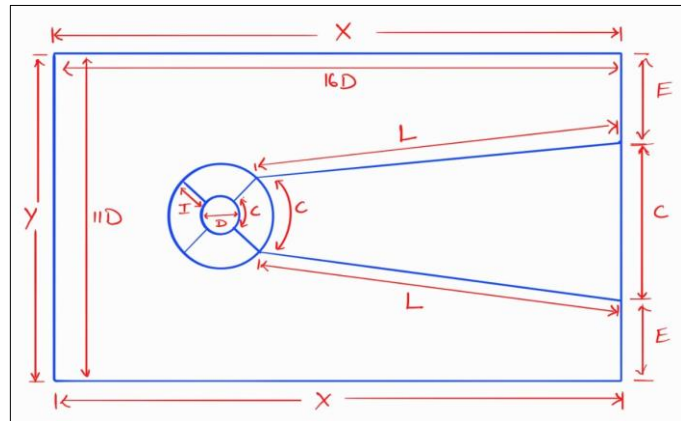
Summer Semester 2025

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Task 2.1 and 2.2

The value of $D = 1$ is used in all the meshes that have been generated. Here letters were assigned to each side in a particular manner because it is easy to explain or for easy understanding of how many points were used on that line or curve to generate the mesh. In this task we have created 3 meshes for same domain. The finest meshes has 1,89,740 nodes and 1,10,045 control volumes, medium with 94,752 nodes and 55,025 control volumes and coarse with 47,760 nodes and 27,861 control volumes.



By looking at 2D picture attached we know its dimensions in x and y direction and the entire surface is extruded in z direction with $3.14 D$ with a single layer.

Table: Number of points taken on surface lines and curves

	n I	n C	n L	n E	n Y	n X
Coarse	35	72	139	14	17	25
Medium	50	100	200	20	24	35
Fine	71	141	284	28	33	47

We have chosen these values as we want to increase the number of cells by factor of 2 as it gets refined i.e., making the mesh finer. First, we have tested the fine with checkMesh command to check its skewness and aspect ratio and if any error occurred, then ran it to check whether iterations were running or not. Then made the finer meshes. Here we made 3 different meshes to study grid convergence. In the above 3 mesh files there is structured and un structured mesh, around and behind cylinder it is structured and remaining part is un structured.

Task 2.3

Table: Pressure and velocity boundary conditions

	Inlet	Outlet	Cylinder	Top/ bottom	Front/ Back
U	1,0,0	zeroGradient	noSlip	1,0,0	empty
p	zeroGradient	zeroGradient	zeroGradient	zeroGradient	empty

Inlet U: At the inlet, assigning a fixed value $U = (1,0,0)$ defines a uniform velocity of the fluid in x-direction that enters the domain. By giving this fixed value at the inlet, we are telling the simulation how fluid enters the domain with fixed value and with preferred direction

p: At the inlet, assigning a zero Gradient condition (Neumann boundary condition) ensures that the change in pressure across the inlet is zero, which will make sure that the pressure is uniform across the inlet with no variation in pressure in the direction of the flow from the inlet into the computational domain which helps in smooth flow across the inlet.

Outlet:

U: At the outlet, assigning a zero Gradient condition (Neumann boundary condition) allows the velocity to be the same as the value across the outlet, which helps in not creating any non-physical and lets the velocities at the outlet leave the domain without any changes.

p: At the outlet, assigning a zero Gradient condition (Neumann boundary condition) makes sure that the flow is fully developed inside the domain and by doing so, we are not fixating the pressure to a certain values, which return may cause adverse pressure gradients and the flow will re-enter the domain and cause inaccurate results, so by keeping the zero gradient at the outlet makes sure that the fluid leaves the domain without any abrupt changes across the outlet.

Wall (cylinder):

U: At the wall(cylinder), assigning a no-Slip condition allows the velocity to be zero at the surface of the cylinder and causes velocity gradients over the cylinder, which also contribute to drag, because this interaction between the fluid and solid is the main reason for the skin friction drag. If the surface of the cylinder is infinitely smooth, the fluid just slips over the surface of the cylinder and there would be no skin friction drag, which is unrealistic. To make our simulation like real world scenario, there should be no slip condition over the cylinder.

p: The gradient normal to the wall of all scalar quantities is equal to zero (ref-CFD 1-L13-pg no-27)

Top and bottom:

U: At the top and bottom, we are giving a velocity, same as inlet condition, which in turn will make sure the flow is fully developed at the top and bottom as well as internal domain and this is eliminate any influence on the cylinder by the top and bottom because there's no relative velocity between the fluid and top and bottom which in turn will eliminate the shear stress which is caused by the velocity gradients formed there.

p: As the top and bottom are moving, with same speed as the flow, it will be like that of Couette flow with two moving walls, placing all the variables in NS equations only at the top and bottom will result in dp/dx and dp/dy to be zero. (ref-Aerodynamics-L3-pg no-30)

Front and back:

U: At the front and back, there are no cells between them, because we are doing a 2D simulation for this setup, but as openfoam only takes 3D simulations, we extruded the domain by pi and assigned empty to the front and back which in turn, it will not affect the setup by any means.

p: Same goes for pressure as well.

K-Omega SST:

	Inlet	Outlet	Cylinder	Top/ bottom	Front/ back
U & p	Same boundary conditions as Laminar				
Turbulent-kinetic energy-K	1e-6	zeroGradient	0	1e-6	empty
Specific dissipation rate-Omega	5	zeroGradient	5	5	empty
nuT (turb. viscosity)	calculated	calculated	0	calculated	empty

(ref-FoamTutorials/incompressible/simpleFoam/pitzDaily and motorBike)

Inlet, Top and bottom:

U & p; Same boundary conditions as Laminar

K: The value for k is taken from (<https://turbmodels.larc.nasa.gov/sst.html>), where it explicitly mentioned, where inlet as well as top and bottom should have specified conditions because turbulent kinetic energy is dependent on the inlet velocity, as there is same velocity of fluid in inlet as well as Top and bottom, same conditions can be applied to inlet as well as top and bottom

Omega: The value for omega is taken from (<https://turbmodels.larc.nasa.gov/sst.html>), where it explicitly mentioned, where L is the scale of the aerodynamic body of interest. Same as k, omega also should have specified conditions at the inlet, but as the top and bottom is also moving with same velocity, same boundary conditions can also be applied for top and bottom like inlet

nuT; The value of turbulent viscosity is computed directly by the formula $\nu_t = C_\mu \frac{k}{\omega}$, where $C_\mu=0.09$, that is why it is given 'calculated' condition.

Outlet:

k: Zero gradient condition is given at the outlet, to ensure that the values are not affected at the outlet, which also tells that the fluid is carrying the same value across the outlet, without effecting the simulation.

omega: Zero gradient condition is given at the outlet, to ensure that the values are not affected at the outlet, which also tells that the fluid is carrying the same value across the outlet, without effecting the simulation.

nuT; The value of turbulent viscosity is computed directly by the formula $\nu_t = C_\mu \frac{k}{\omega}$, where $C_\mu=0.09$, that is why it is given 'calculated' condition. (ref-L9-pg no17, 28)

Wall (cylinder):

k: The value for k at the wall(cylinder) is taken as zero, because the velocity is zero at the surface of the cylinder due to no slip condition, which in turn will result in zero kinetic energy.

omega: The value for omega at the wall is non zero at the cylinder, as the free-stream flow is laminar with no free-stream turbulence, the flow at the surface of the cylinder will experience a very high shear stress due to velocity gradients, this will result in formation of eddies and vortices even though the inlet stream itself is laminar, this results in non-zero omega as the turbulent kinetic energy is converted to thermal energy.

nuT: The value of turbulent viscosity is zero as the value of k is zero, which can be directly shown by the formula $\nu_t = C_\mu \frac{k}{\omega}$.

Front and back: Empty as for 2D RANS.

Spalart-Allmaras:

	Inlet	Outlet	Cylinder	Top/ bottom	Front/ back
U & p	Same boundary conditions as Laminar				
Turbulent viscosity nuT	0.0000539	0.0000539	0	0.0000539	empty
Modified turbulent viscosity nuTilda	0.000769	0.000769	0	0.000769	empty

(ref-FoamTutorials/incompressible/simpleFoam/airFoil2D)

Inlet, Top and bottom:

U & p: Same boundary conditions as Laminar

nuT: The value of turbulent viscosity is taken from the reference(<https://turbmodels.larc.nasa.gov/spalart.html#sa>), where $\nu_\infty = 2.564e-4$ (calculated from $Re_D=3900$) and it is assigned a freestream value, because the freestream boundary condition will switch between fixed (free stream) value and zero gradient based on the sign of the flux (ref <https://doc.openfoam.com/2306/tools/processing/boundaryconditions/rtm/derived/inletOutlet/freestream/>)

nuTilda: The value of turbulent viscosity is taken from reference (<https://turbmodels.larc.nasa.gov/spalart.html#sa>), where $\nu_\infty = 2.564e-4$ (calculated from $Re_D=3900$) and it is assigned a freestream value same as nuT (turbulent viscosity)

Outlet:

U & p: Same boundary conditions as Laminar

nuT: Same boundary conditions as inlet as the freestream boundary condition will switch between fixed (free stream) value and zero gradient based on the sign of the flux

nuTilda: Same boundary conditions as inlet as the freestream boundary condition will switch between fixed (free stream) value and zero gradient based on the sign of the flux

Wall (cylinder):

U & p: Same boundary conditions as Laminar

nuT: The value of turbulent viscosity is zero at the surface of the cylinder, because there is no velocity at the surface of the cylinder due to the no slip condition, which will cause no turbulent eddies to occur at the wall itself, this can also be portrayed, as $k=0$, $\nu_t=0$

nuTilda: The value of modified turbulent viscosity is also zero, as $\nu_t = \hat{\nu} f_{v1}$, and this values also mentioned in the above reference(<https://turbmodels.larc.nasa.gov/spalart.html#sa>)

Front and back: Empty as for 2D RANS.

Task 2.4

Performing URANS simulation

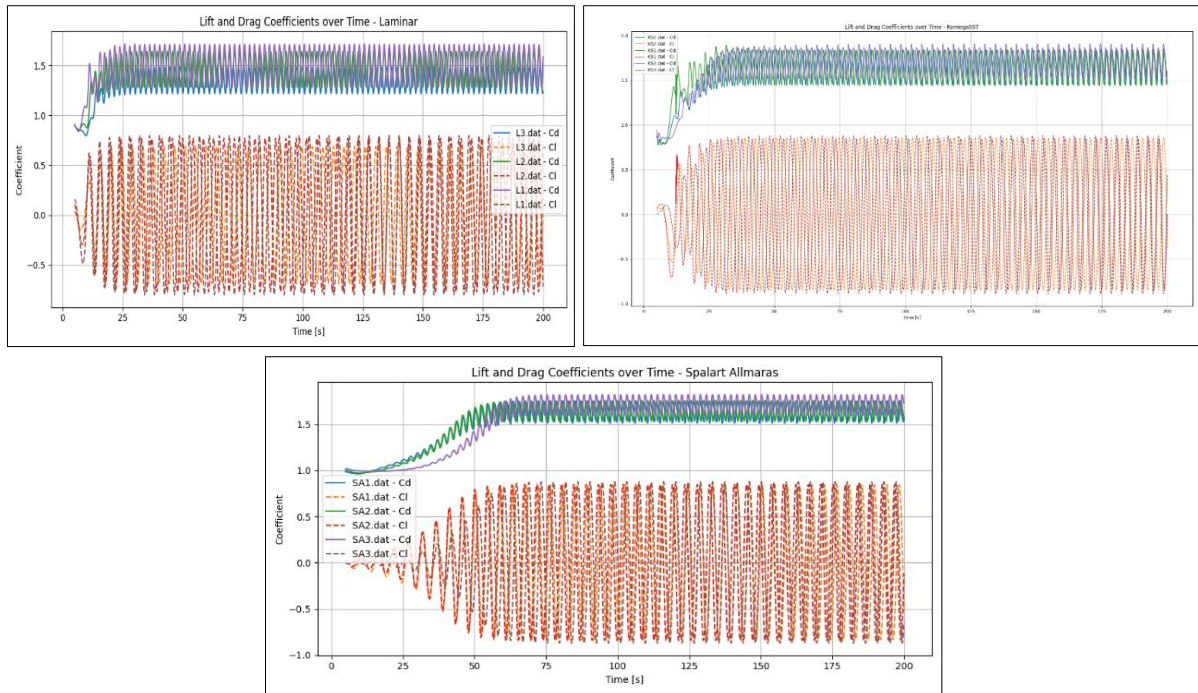
1. As the boundary conditions are already chosen, we just need to give them in the 0 directory in the simulation setup for respective models (K-omegaSST, Spalart Allmaras, and Laminar (no-turbulence model)).
2. After creating the meshes with three different control volumes, we have to import the mesh from gmsk to openfoam and after importing, changes the patch of the cylinder to wall and as well as change the patch of front and back to empty in polymesh which is in constant directory.
3. Specify the kinematic viscosity in the physical properties and specify the RANS model with simulation type with laminar, KomegaSST and SpalartAllmaras
4. In the controlDict, specify the timestep, which will differ for respective meshes, and input the force coefficients code at the end by using the inbuilt functions in openfoam
5. The simulation gets diverged when used gauss linear in fvSchemes for divergent schemes, this happens because the gauss linear is unbounded and central differencing scheme, from CFD1, we know that CDS is unstable for advection dominant flows, that is why “bounded gauss upwind” is used for laminar simulation and Gauss limitedLinear 1 is used for KomegaSST and Spalart Allmaras, which uses CDS for interpolation and upwind for stability as it’s a advection dominant flow. For time derivative schemes, we used “backward” because it is second order accurate. Below attached are the images of simulations that got diverge (ref: <https://doc.cfd.direct/openfoam/user-guide-v10/fvschemes>)
6. For solver in fvSolution, we used pimpleFoam because it is a transient solver for incompressible, turbulent flow of Newtonian fluids and it is combination of SIMPLE and PISO algorithm, where it utilizes the robustness of SIMPLE and transient feature from PISO.
7. The same setup is also used for RANS with respective changes in solver (SIMPLE) and time derivate to be steady state

Lift and drag coefficients for each model for three different meshes:

	Coarse mesh		Medium mesh		Fine mesh	
	Cd	Cl	Cd	Cl	Cd	Cl
KomegaSST	1.7144	0.0142	1.6618	0.0044	1.6565	0.0031
Spalart Allmaras	1.7140	-0.0153	1.6509	-0.0026	1.6462	0.0084
Laminar	1.3565	-0.0129	1.4688	0.0054	1.5077	0.0002

Task 2.5:

By looking at the above images, we can say that the Cd and Cl values of all models (KomegaSST, Spalart Allmaras, Laminar) are converged, as the mean value, fluctuating amplitude and frequency does not change over time. The error percentage between the coarse and finest mesh is (KomegaSST- 3.50%), (Spalart Allmaras-4.12%), (Laminar- 10.03%). We can clearly that the drag coefficient for KomegaSST and Spalart Allmaras are converged, because the value does not change more than a 5% (mentioned in the question) between the highest and lowest resolution grids, but for laminar its more than 5%, but it’s getting decreased when we check the error percentage between the medium and fine mesh, which is for laminar is 2.58%.



Task 2.6:

Comparing models with each other:

By comparing the results between KomegaSST, Spalart Allmaras for the fine mesh, we can see that the Cd values are very close to each other, which has percentage difference on the fine mesh is approximately of 0.628%, while Laminar has around 8 to 9 % difference with KomegaSST and Spalart Allmaras. We think this is because, we are not considering turbulent effects for laminar, as turbulence in the wake region and around the cylinder also accounts for the drag, which we are not taking into consideration, so the drag for laminar should be lowest among the models.

Comparing models with Literature:

By comparing the results with literature 'Numerical study of the flow over a circular cylinder in the near wake at Reynolds number 3900' published by American Institute of Aeronautics and Astronautics (AIAA), which shows the results of Cd as follows:

Table 2 Computed mean quantities for structured and unstructured mesh 2-D simulation in comparison to experimental

	Experiment ¹⁰	Mesh Sets	Structured mesh	Unstructured mesh
$\overline{C_D}$	0.98±0.05	Grid I	1.39	0.60
		Grid II	1.43	0.72
		Grid III	1.42	0.93

By comparing the results of transient 2D simulation (which are also over-predicted) with ours, we found the reason for the over-predicted values of Drag coefficient is that the higher Reynolds stresses and closer location of intense Reynolds stresses are the main cause for the over-prediction of base suction pressure level and resulting higher drag and lift in two-dimensional simulations. Therefore, predicting pressure distribution accurately is crucial to determining one of the important quantities, namely the drag force.

ref(https://www.researchgate.net/publication/261699686_Numerical_Study_of_the_Flow_Over_a_Circular_Cylinder_in_the_Near_Wake_at_Reynolds_Number_3900)

Task 2.7

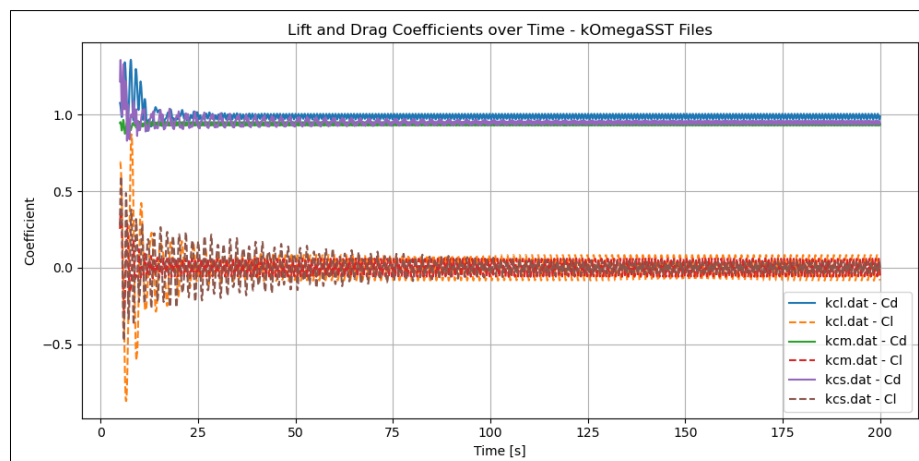
The previous steps were repeated for steady RANS.

Initially we have done setup Steady RANS with 3 different models namely k-Omega SST, Spalart-Allmaras and Laminar, we have selected SIMPLE solver because the simulation is steady state and this solver is good for steady state incompressible flows and we can also use under relaxation factors for the equations.

1. Changes that have been made in the 0, constant and system directories. In the zero file U, p, k, omega, nu, nuTilda are taken same as explained in boundary conditions.
2. In the momentum transport file, changes were made for each type of model taken to simulate. For Spalart-Allmaras simulationType is RANS and model is spalartAllmaras, similarly simulationType is same for kOmegaSST but model is changed to kOmegaSST and for no turbulence model simulation type is laminar.
3. In the physical properties file, viscosityModel is taken as constant, nu(viscosity) value is taken as 2.564e-6 which is 3900 Reynolds number. It is same for all the models.
4. In the controlDict file application type is changed to simpleFoam as we are using SIMPLE solver which is more widely used for RANS application, endTime is fixed to 200 because while doing simulation for coarse mesh we found that result is converging for 120 to 150 time, so that's why we have taken 200-time steps.
5. In the fvSchemes ddt Schemes change to steadyState because simple solver itself is a steady state solver.
6. In the fvSolution, for solvers p, U, k, omega and nuTilda tolerance values are taken from default tutorial files and solver is change to SIMPLE and there are relaxation factors for fields p, U, k, omega.

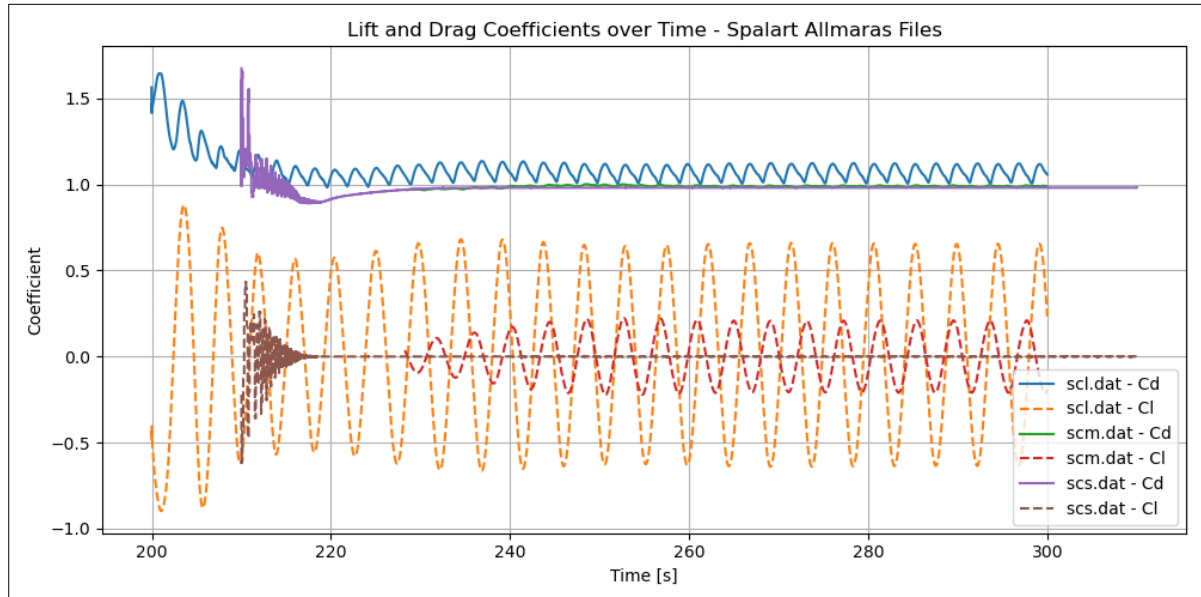
For getting results in RANS, we have considered or taken result from URANS and ran those files for steady state RANS over 200-time steps. Comparison of lift coefficient (Cl) and drag coefficient (Cd) for three different models with different grids is as follows:

K-Omega SST:



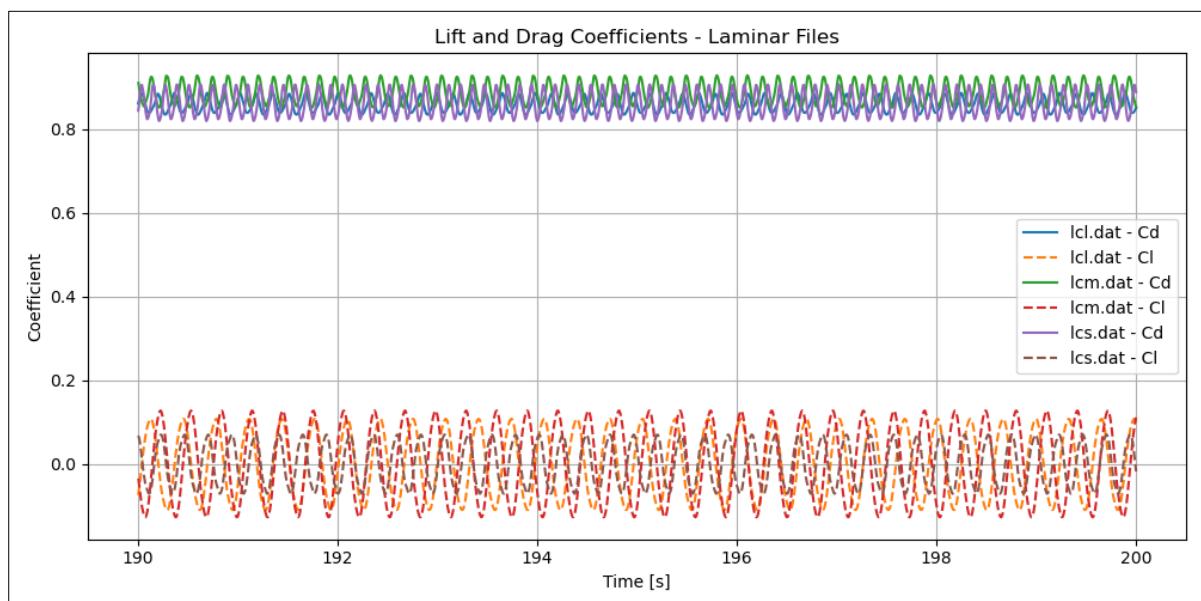
kOmegaSSt	Cd	Cl
Coarse	0.98891	-0.00014
Medium	0.94079	0.00028
Fine	0.94961	0.00053

Spalart-Allmaras:



Spalart-Allmaras	Cd	Cl
Coarse	1.07077	0.00689
Medium	0.99011	-0.00024
Fine	0.98114	-0.00023

Laminar:

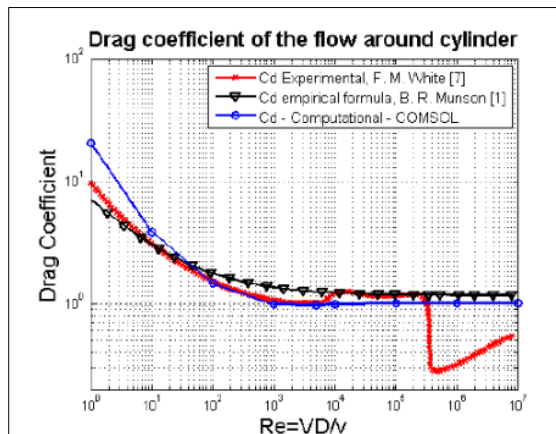


Laminar	Cd	CI
Coarse	0.85766	-0.00068
Medium	0.88578	-0.00022
Fine	0.86077	-0.00014

Comparing the results of different models of same mesh (finest)

Fine Mesh	Cd	CI
K-Omega SST	0.94961	0.00053
Spalart-Allmaras	0.98114	-0.00023
Laminar	0.86077	-0.00014

Here we can observe that for modelled turbulent flows such as kOmegaSST and Spalart-Allmaras have almost the same Cd value and the percentage of difference between them is less than 5%, here we have pressure(p) and velocity(U) fields along with separate fields or model for turbulent kinetic energy(k), turbulent viscosity(nut) and modified viscosity(nuTilda), but for the laminar flow it only has pressure (p) and velocity(U) fields. Due that that reason the Cd value changed more than 5%.



Turbulence models	C_D	θ_{sep}	C_{bp}	L_r/D
$k - \omega$	0.79	95.9	-0.49	2.7
LES D3 run (Breuer, 1989)	1.02	87.4	-0.94	1.37
Experiment (Norberg, 1987)	0.98 ± 0.05	85 ± 2.0	-0.9 ± 0.05	1.33 ± 0.2

(Ref:

<https://www.researchgate.net/publication/287587535>)

(Fig Ref:

<https://www.researchgate.net/publication/274309261>)

In the above attached reference, we can see that k-omega model is used in steady RANS and they got result of drag coefficient Cd 0.78 and given value can also be around $0.98 \pm 5\%$ which is matching with our results in steady RANS.

The above images from the online papers were done using RANS where we can see in general the coefficient of drag (Cd) for Reynolds number from 1000 to 10000 is a bit less than 1 (approx. 0.9) and we got the Cd values 0.95 on average and this matches with images shown above. We think kOmegaSST and Spalart-Allmaras yields more accurate results than Laminar because as we discussed above, they got separate models apart from both p and U equations which helps to capture the flow fields more accurate and less error in the results.

Task 2.8

We can see that there is a change in Cd and CI values for all the 3 models in RANS and URANS because of the solvers that we are using in RANS and URANS are different we are using SIMPLE solver which is used for steady state flows where we don't consider the change of flow properties with respect to time i.e., dt value does not matter because there is no partial differentiation term with respect to time where as in the Unsteady RANS we use PIMPLE where we consider changes of flow properties with

respect to time and we choose time step Δt in such a manner that we maintain courant number less than or equal to 1, which captures all the information at each and every time step.

Comparison of RANS and URANS lift and drag for the finest mesh:

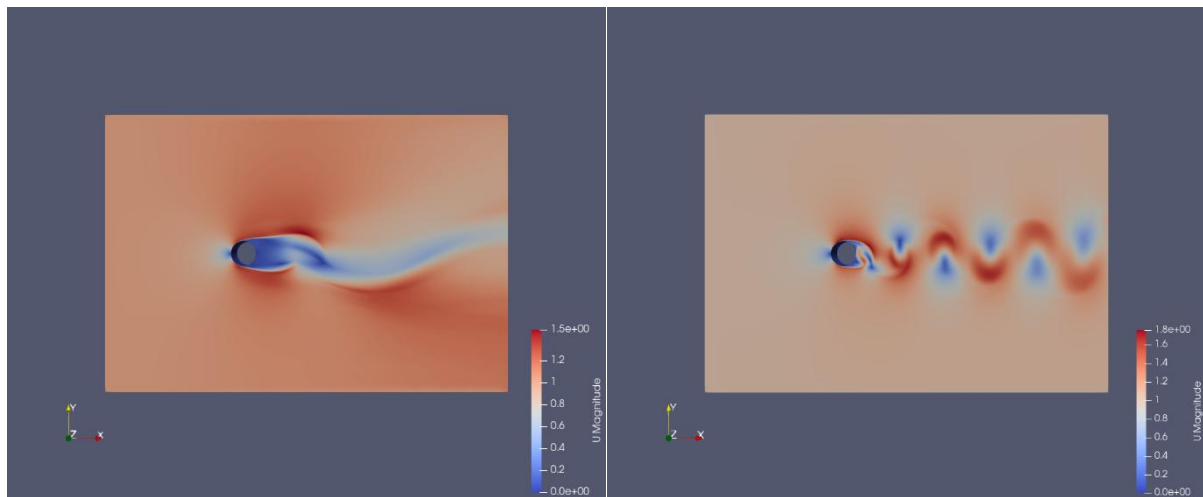
Model	RANS		URANS	
	Cd	Cl	Cd	Cl
K-Omega SST	0.94961	0.00053	1.6565	0.0031
Spalart-Allmaras	0.98114	-0.00023	1.6462	0.0084
Laminar	0.86077	-0.00014	1.5077	0.0002

Task 2.9

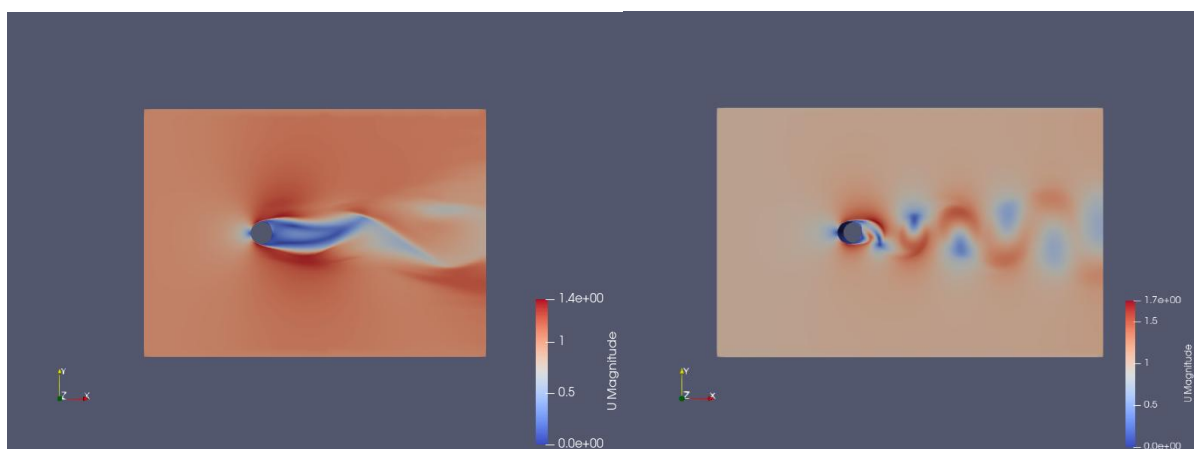
RANS

URANS

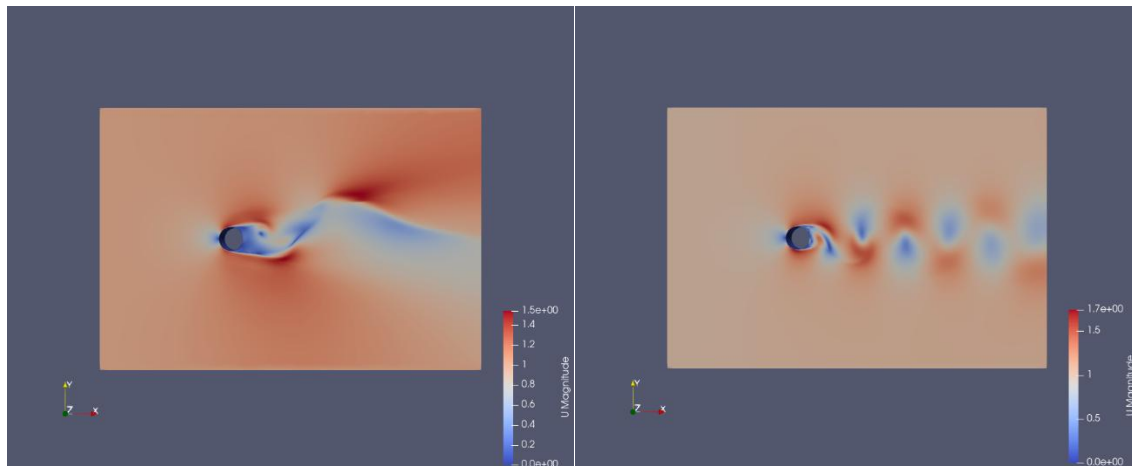
K-Omega SST velocity magnitude



Spalart-Allmaras velocity U-Magnitude



Laminar (no model) velocity U-magnitude



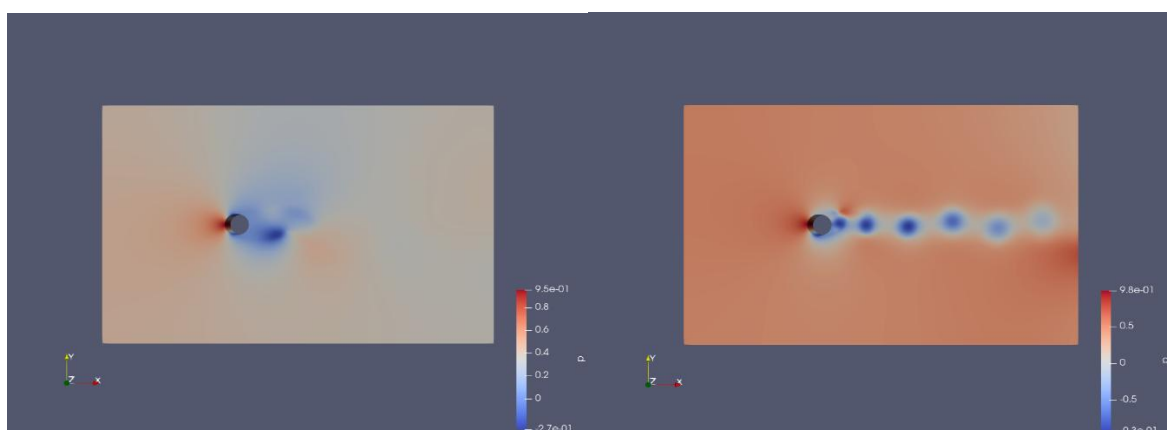
The above contour plots are the velocity U-Magnitude of both RANS and URANS for all the three models namely kOmegaSST, Spalart-Allmaras and Laminar, where we can see that the maximum U-magnitude for RANS varies from 1.4e+00 to 1.5e+00 where as for URANS it varies from 1.7e+00 to 1.8e+00.

When we compare our drag coefficient results with literature or papers available on open source, we found that for RANS the Cd value is around 0.98 and for URANS it is around 1.6, but when we compare RANS to URANS there is drastic change because of the solvers we are using and also consideration of changes with respect to time, where we include it in the equations that we use, we know that the Reynolds number provided is 3900 where it falls into transient flows which is unsteady in nature so URANS yields more accurate results in terms of unsteady flows and also we use PIMPLE solver which is good for both steady and unsteady flows as it is the combination of both PISO and SIMPLE.

RANS

URANS

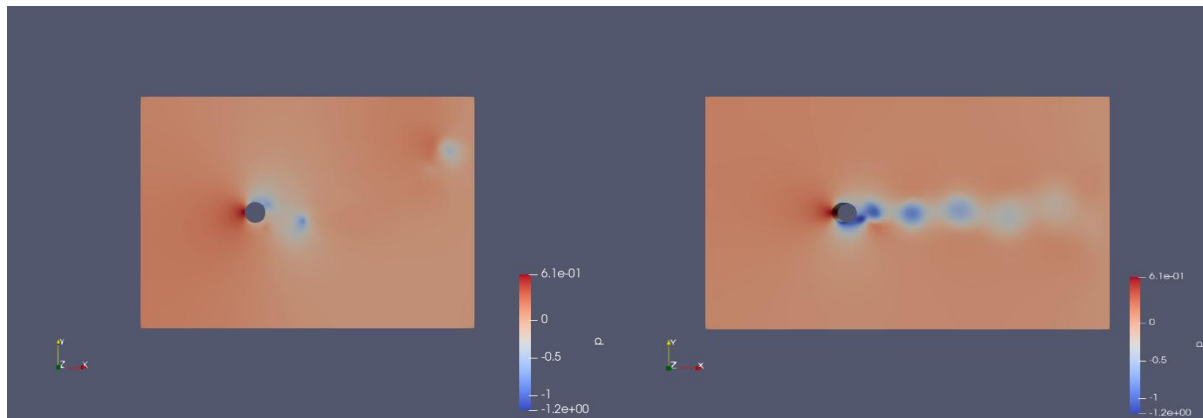
K-Omega SST pressure p-contour



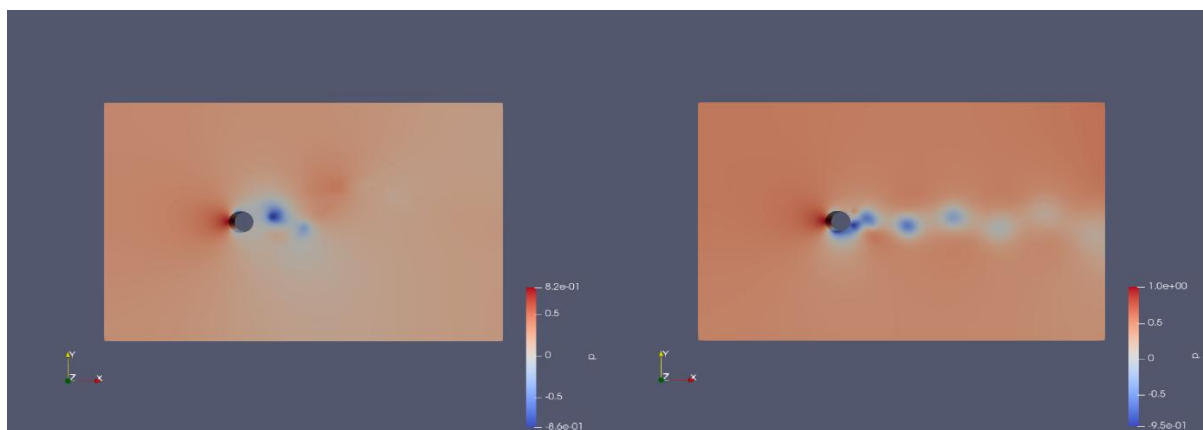
RANS

URANS

Spalart-Allmaras pressure p-contour



Laminar (no model) pressure p-contour



Task 2.10

Initial setup: LES was run in parallel for 200 seconds to allow flow to develop. The 3D simulation used the mesh provided and the schemes used were as mentioned below. The reference used is the default case from OpenFOAM 'channel395' and 'pitzDaily'.

fvSchemes:

LES requires schemes for time, gradient, divergence. For time, the scheme 'backward' was used which is transient, second order implicit, unbounded. The gradient scheme used was 'Gauss linear' which interpolates schemes from cell face to cell centre (4.5.2 in Ref). We had multiple runs with different divergence schemes. In the first run we used Gauss Linear which is CDS. Phi denotes the volumetric

flux and the term $\text{div}(\phi, U)$ is thus the advective term. From our CFD I lectures, it was clear that convective term requires different treatment. But we tried first with the Gauss linear to see the output. As shown in the image, CDS gives visible streaks which clearly indicate the issue. The literature suggests LUST (blended 75% linear/ 25% linear Upwind scheme) which was then chosen which gives satisfactory results. The snGrad scheme used was corrected to account for real world non orthogonal mesh. The Laplacian scheme used is default Gauss linear. <https://doc.cfd.direct/openfoam/user-guide-v10/fvschemes>

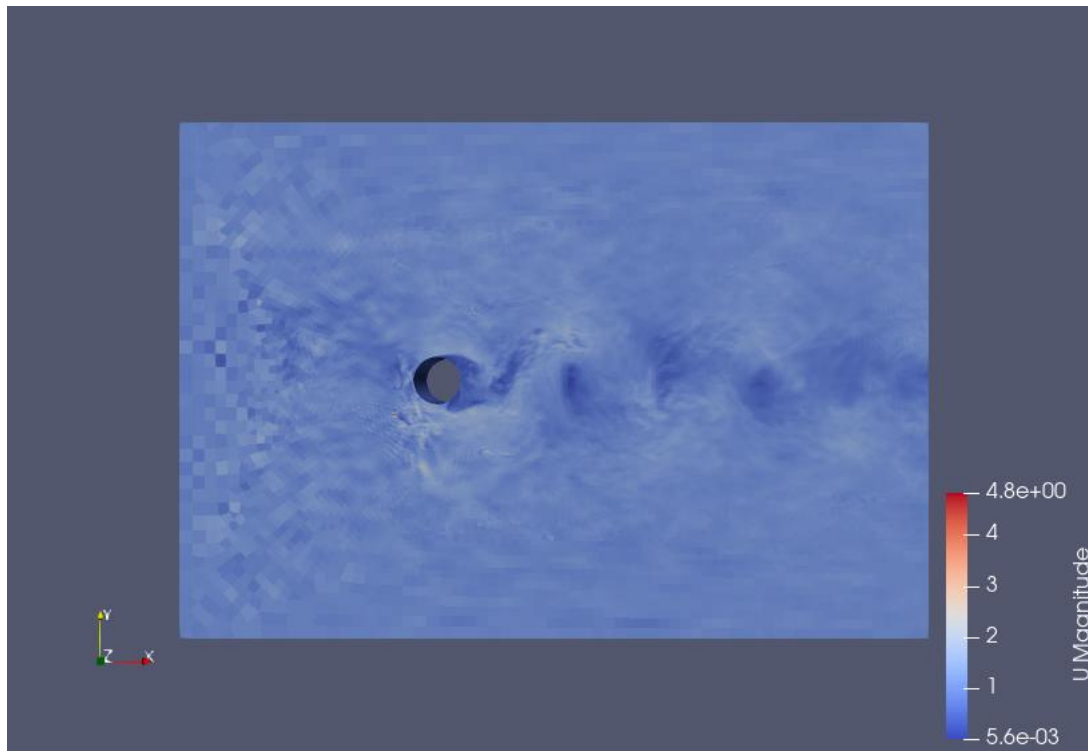


Figure: Effect of wrong scheme for advective terms.

fvSolution:

For pressure GAMG and for velocity smoothSolver is used. GAMG was used as it is faster because it calculates the initial guess using an internal coarse mesh and then projects it to the actual mesh (4.6.1.4).

Pimple foam was the solver used with nCorrectors 2 which solves pressure equation 2 times per iteration. nOuterCorrectors was 2 which is no. of times the complete system is solved. It was observed that this step increased the time. nOrthogonal correctors 0 and 1 were tried with no visible difference hence left 0 for further runs.

Boundary conditions:

The pressure and velocity conditions are same as URANS/RANS, with the exception that front and back are now set to cyclic instead of empty since LES is a 3D simulation.

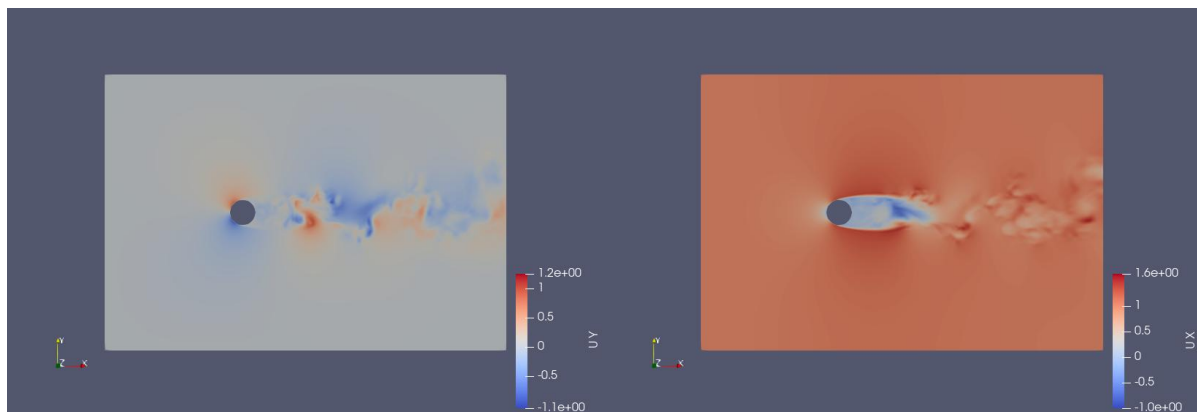
LES additionally requires boundary condition for ν_t . The table illustrates at each patch the conditions for ν_t . The filtered NS are (Ref Lecture LES 1 slide 31):

$$\frac{\partial \bar{u}_j}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_i \bar{u}_j) = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x} (\nu + \nu_t) \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right)$$

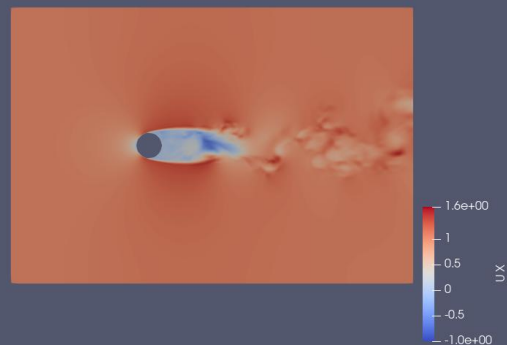
Where ν_t is the eddy viscosity and ν is the molecular viscosity. The eddy viscosity is modeled using the WALE model and therefore we can put the value of ν_t to be zero in the free stream since it is considered turbulence free inlet and, in the equation, eddy viscosity appears along with molecular viscosity in the numerator and hence it can be put 0 unlike some other procedures like RANS where we must provide a non-zero small value. At the cylinder, both Fixed value 0 and zero gradient conditions were tested with minimal changes in the flow. The top and bottom have same condition as the inlet as they are free stream. The outlet is zero gradient in the hope that the flow is fully developed. Front and back are cyclic like LAB 1 channel case. The coefficient for the WALE model were the default coefficients C_e , C_k , C_w to be 1.048, 0.094, 0.325. <https://www.openfoam.com/documentation/guides/latest/doc/guide-turbulence-les-wale.html> .

Front/ Back	Inlet	Outlet	Cylinder	Top/ Bottom
Cyclic	0	Zero Gradient	Zero Gradient/ Fixed 0	0

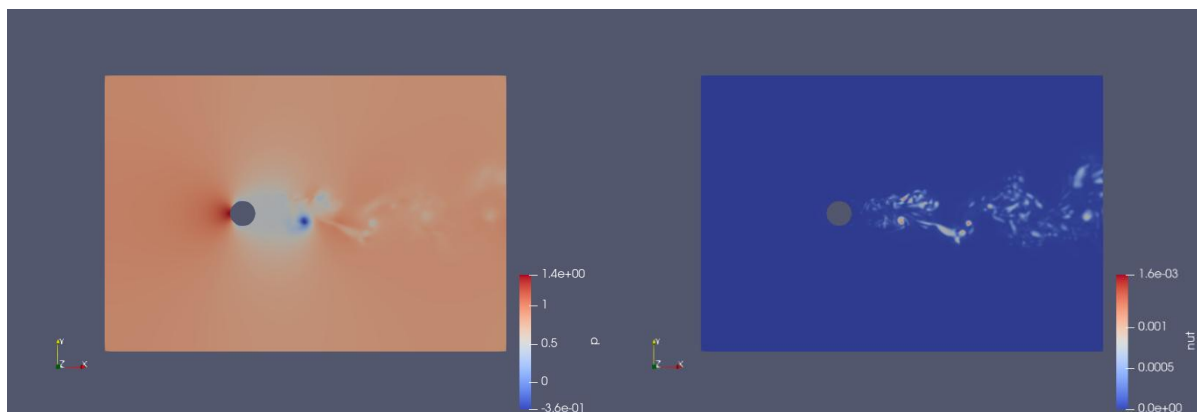
Velocity Ux



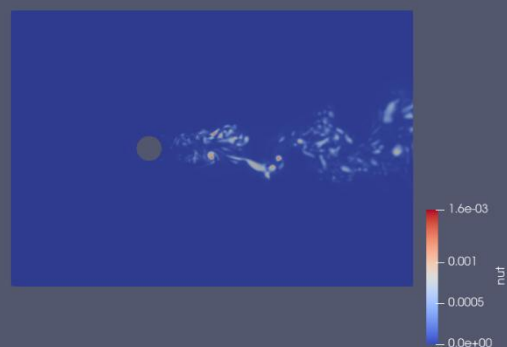
Velocity Uy



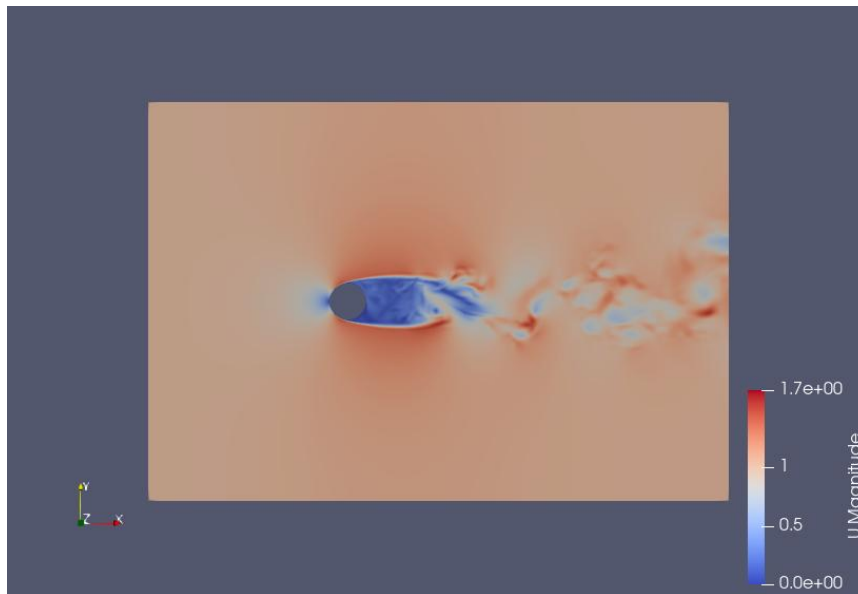
Pressure



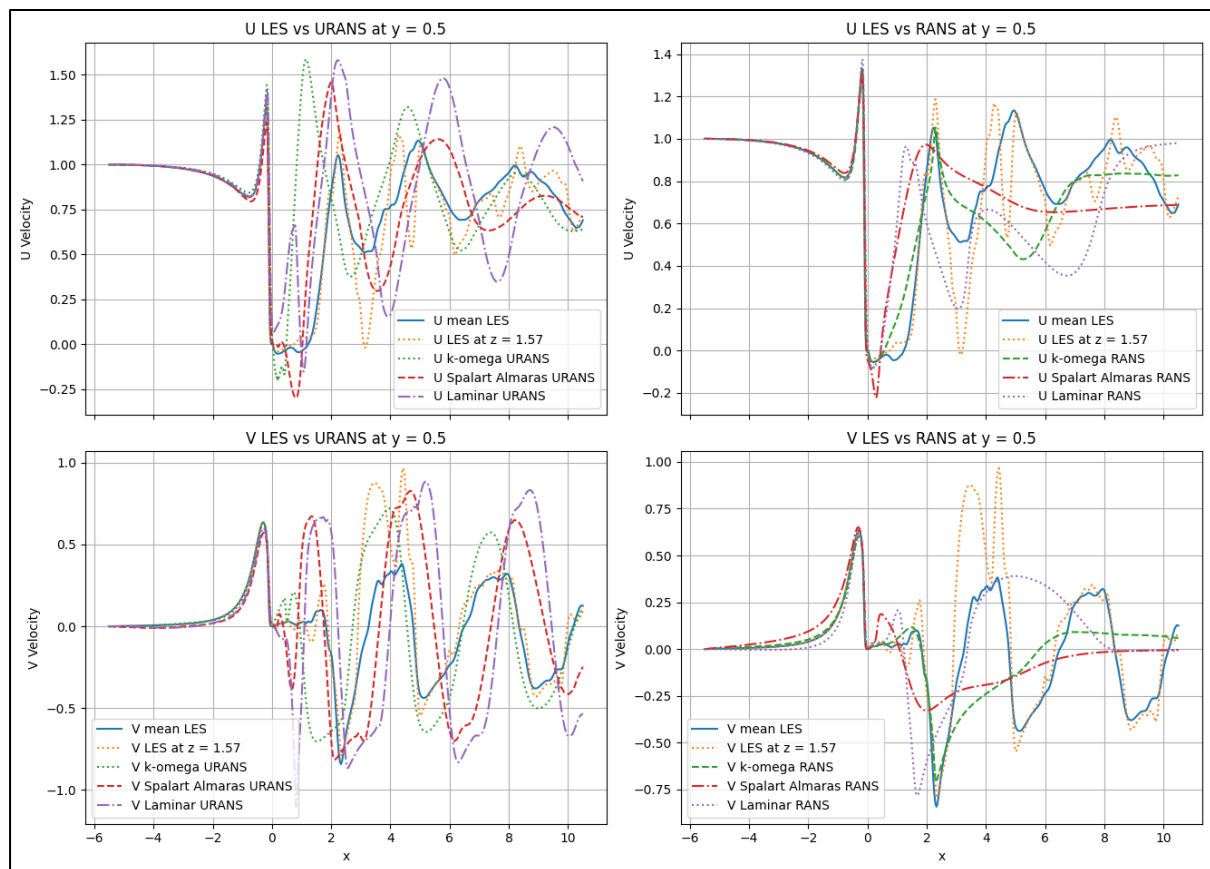
NuT



Velocity magnitude:



Task 2.11



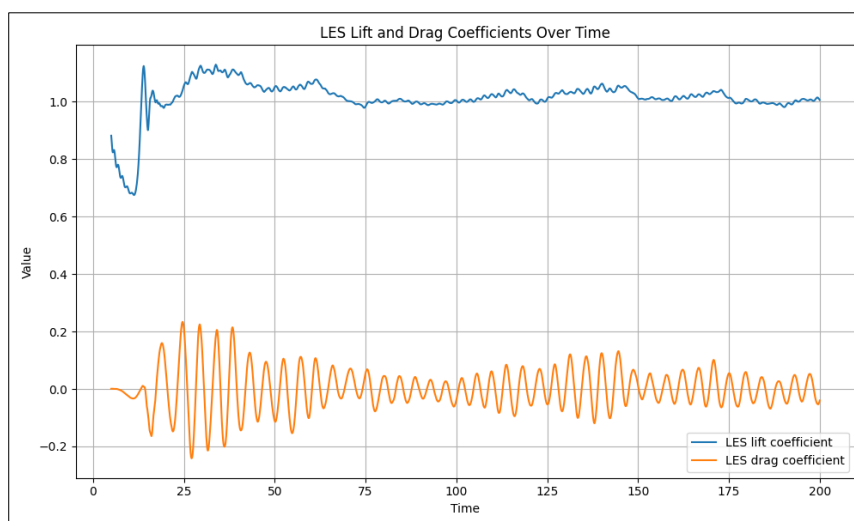
The values are plotted at $y = 0.5$ which is just on the top surface of the cylinder. The LES data is Z direction averaged. We observe that the initial flow for all the models is similar before the cylinder.

The U velocity has distinct peaks in LES which are seen in most models but with positions shifted. We observe that the URANS and RANS models are showing typically higher values than LES. An instantaneous LES velocity is also plotted at the domain centre. We observe that the K Omega model (green dotted line) is similar to LES plot regarding fluctuations. The no model laminar plot is much different in nature and the flow visualization also confirm its unphysical nature. We also observe that the shedding has started earlier for URANS/ RANS models.

Task 2.12

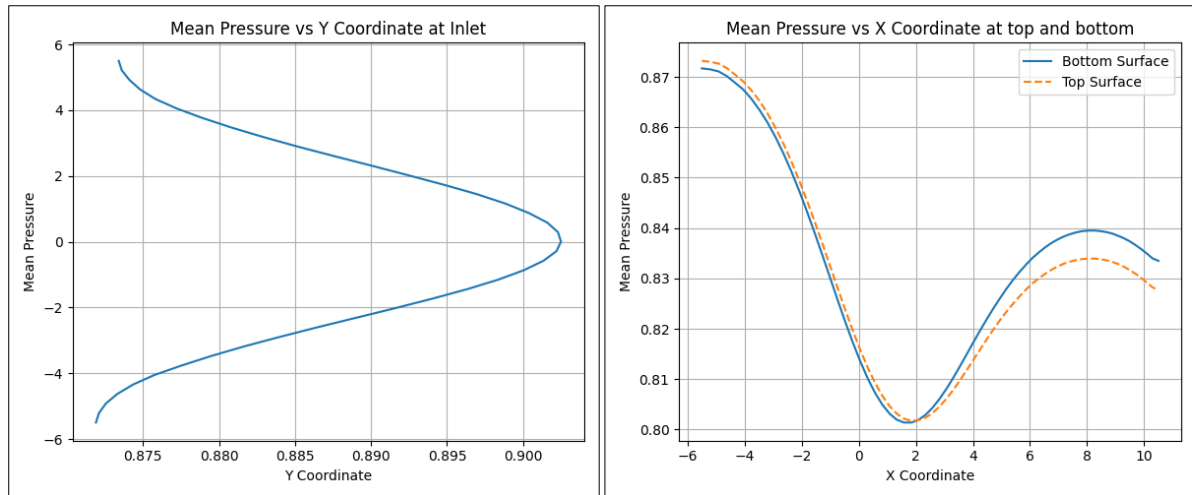
Discarding transients, the mean C_d for LES was 1.027 and C_l was 0.001. These results closely match literature such as Shim et. al. (10.2514/6.2009-4160) and B. N. Rajan et al (ISSN 1735-3572). Kravchenko and Moin reported a C_d of 1.04 and the WALE simulation by Ouvrard et. al. reported a C_d of 1.02.

The Laminar case gives an unphysical output where the flow produces no eddies due to absence of turbulent model. The RANS models although having a close enough values of C_d to LES produced no vortex shedding which is also unphysical. We have realised that the vortex shedding is a time dependent phenomenon and a steady state solver does not capture it. Therefore, the URANS although the C_d values being different were capturing the Karman vortex phenomenon. Also, the vortex peaks match somewhat with the LES data as seen in Task 2.11



Task 2.13

The plot is for mean pressure with Z averaging. From the first plot, we see that the pressure increases near the domain centre which is behind the cylinder. This indicates that the cylinder is affecting pressure behind it, which is not advisable with regards to domain length, meaning the domain is not long enough behind the cylinder. The top and bottom plots show this effect too and are affected by the presence of cylinder which is at 0 coordinate. In best case, there should be no effect on the top, bottom, inlet boundaries. Here we also understood the elliptic nature of pressure and how it affects the entire domain, not just the flow direction.



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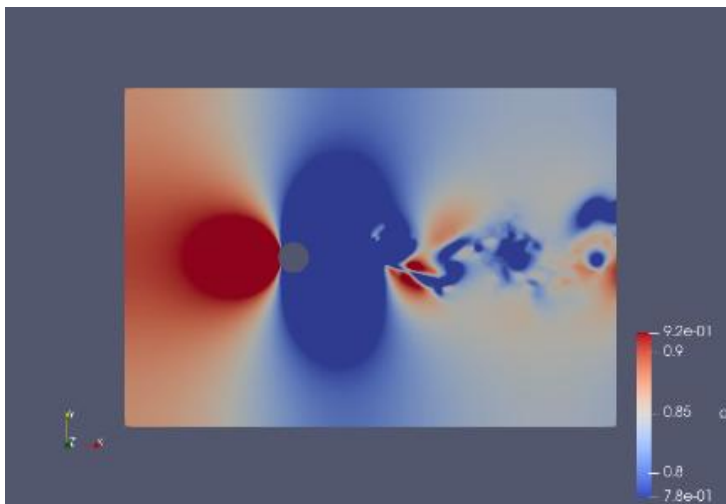
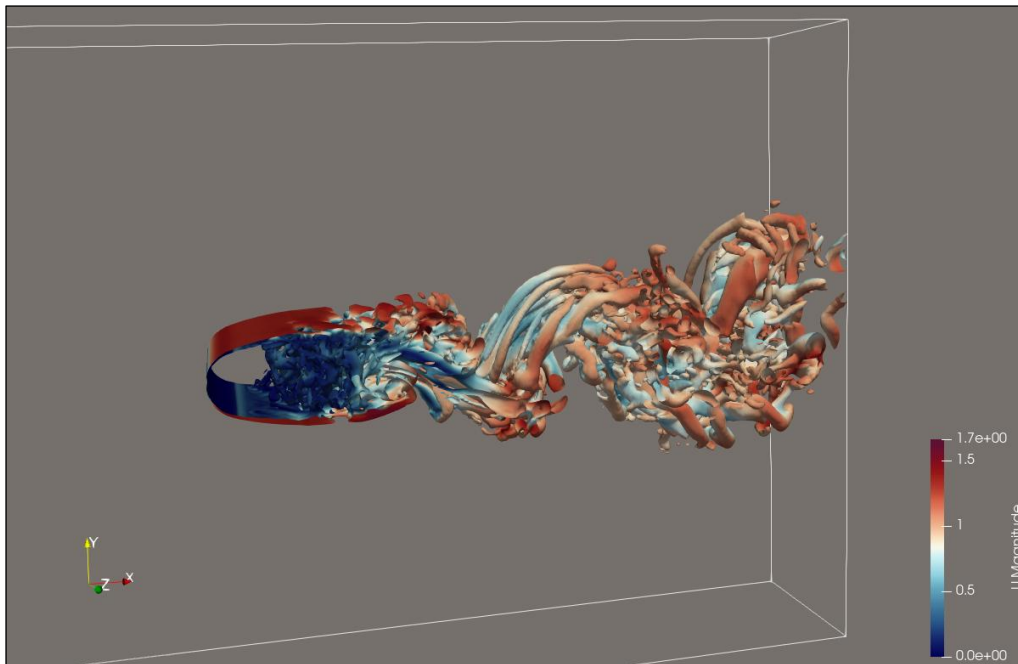


Figure: Rescaled pressure contour to show effect on domain

Task 2.14

Q surfaces are an indicator of turbulent flow structures. Eddy is defined as the region with positive second invariant, Q such that the pressure should be lower than the ambient. Q represents the local balance between shear strain rate and vorticity magnitude. Value of Q is positive, which correctly identifies surfaces. The value chosen for Q was 0.5 (Jeong-Hussein 1995)

We can see that the turbulent structures are correctly identified. Smaller structures are identified on further decreasing the value of Q , whereas large values only identify bigger vortices.



Contributions:

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3,4,5,6: Mohan Kumar Meesala

10,11,12,13,14: Mihir Ajit Lonkar